



Jérôme Eertmans

Research intern at KDDI Research, Inc. | Postdoctoral researcher and Teaching Assistant at UCLouvain.

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Education

UCLouvain, Ph.D. in Electrical engineering Sept 2021 – July 2026
Thesis title: Differentiable Ray Tracing for Radio Propagation Modeling. Available [here](#).

UCLouvain, M.Sc. in Electro-mechanical engineering Sept 2019 – June 2021
Majored in mechatronics.

- I undertook a variety of course topics: telecommunications, open source software, numerical simulations, digital design of ICs, machine learning, mining pattern in data, etc., for a total of 112% workload.
- Graduated Magna Cum Laude.

Experience

Research Intern, KDDI Research, Inc. – Fujimino, Saitama, Japan July 2026 – present
Working on machine learning for indoor radio propagation modeling, within the radio propagation and devices group.

Teaching Assistant, UCLouvain – Louvain-la-Neuve, Belgium Sept 2021 – present

- Teaching duties (practical sessions, evaluation, ...): see list [here](#).
- Research on my Ph.D. subject under the supervision of Profs. Claude Oestges and Laurent Jacques.

Visiting Researcher, University of Bologna – Bologna, Italy Sept 2024 – Dec 2024
Worked with Profs. Vittorio Degli-Esposti and Enrico Maria Vitucci on Machine Learning applied to Ray Tracing for Radio Propagation, as part of our 'Generative Path Sampling' research project. Led to two conference papers and one journal paper.

For additional experiences, see my [LinkedIn profile](#).

Languages

French: Native

English: B2+/C1

Dutch: B1

Italian: A2

Japanese: Basics

Projects

DiffeRT

Python package with backend written in Rust.

- Built on top of [JAX](#) for high-performance and differentiable code
- Thoroughly tested and well documented
- Fully open access, along with its 2D variant [DiffeRT2D](#)

Manim Slides

Presentation tool I developed to produce higher-quality presentations at conferences.

- Most of my presentations are created using this tool and can be viewed on [my personal website](#)
- Used by other researchers to disseminate their research, with examples available in the [documentation](#)

More projects

You may find an extended list of my projects and contributions on my [GitHub profile](#) or on the [projects page](#) of my website.

Publications

- Transform-Invariant Generative Ray Path Sampling for Efficient Radio Propagation Modeling** Mar 2026
Jérôme Eertmans, Enrico Maria Vitucci, Vittorio Degli-Esposti, Nicola Di Cicco, Laurent Jacques, Claude Oestges
arxiv.org/abs/2603.01655 (arXiv preprint, submitted to npj Wireless Technology)
- Fast, Differentiable, GPU-Accelerated Ray Tracing for Multiple Diffraction and Reflection Paths** Oct 2025
Jérôme Eertmans, Sophie Lequeu, Benoît Legat, Laurent Jacques, Claude Oestges
arxiv.org/abs/2410.14535 (arXiv preprint, accepted at EuCAP 2026)
- Towards Generative Ray Path Sampling for Faster Point-to-Point Ray Tracing** Oct 2024
Jérôme Eertmans, Nicola Di Cicco, Claude Oestges, Laurent Jacques, Enrico Maria Vitucci, Vittorio Degli-Esposti
[10.1109/ICMLCN64995.2025.11140249](https://doi.org/10.1109/ICMLCN64995.2025.11140249) (2025 IEEE International Conference on Machine Learning for Communication and Networking (ICMLCN))
- Comparing Differentiable and Dynamic Ray Tracing - Introducing the Multipath Lifetime Map** Oct 2024
Jérôme Eertmans, Enrico Maria Vitucci, Vittorio Degli-Esposti, Laurent Jacques, Claude Oestges
[10.23919/EuCAP63536.2025.10999736](https://doi.org/10.23919/EuCAP63536.2025.10999736) (2025 19th European Conference on Antennas and Propagation (EuCAP))
- Fully Differentiable Ray Tracing via Discontinuity Smoothing for Radio Network Optimization** Mar 2024
Jérôme Eertmans, Laurent Jacques, Claude Oestges
[10.23919/EuCAP60739.2024.10501570](https://doi.org/10.23919/EuCAP60739.2024.10501570) (2024 18th European Conference on Antennas and Propagation (EuCAP))
- DiffeRT2d - A Differentiable Ray Tracing Python Framework for Radio Propagation** June 2024
Jérôme Eertmans, Claude Oestges, Laurent Jacques
[10.21105/joss.06915](https://doi.org/10.21105/joss.06915) (Journal of Open Source Software)

Min-Path-Tracing - A Diffraction Aware Alternative to Image Method in Ray Tracing Jérôme Eertmans, Claude Oestges, Laurent Jacques 10.23919/EuCAP57121.2023.10132934 (2023 17th European Conference on Antennas and Propagation (EuCAP))	Mar 2023
Manim Slides - A Python package for presenting Manim content anywhere Jérôme Eertmans 10.21105/jose.00206 (Journal of Open Source Education)	Aug 2023

Grants and Awards

Research internship at KDDI Research, Inc. – Fujimino, Saitama, Japan FRS FNRS <i>Accomplishment of a stay abroad.</i>	Sept 2024 – Dec 2024
Best Propagation Paper Award – Dublin, Ireland EurAAP - EuCAP 2026.	Apr 2026
TICRA-EurAAP Travel Grant – Dublin, Ireland EuCAP 2026 travel grant.	Apr 2026
Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (FNRS) – Bologna, Italy FRS FNRS <i>Accomplishment of a stay abroad.</i>	Sept 2024 – Dec 2024
Long-term Scientific Stay in Pr. Vittorio Degli-Esposti's team (WBI) – Bologna, Italy WBI <i>Bourses d'Excellence WORLD.</i>	Sept 2024 – Dec 2024
STSM CA20120 - Visit to Pr. Vittorio Degli-Esposti – Cesena, Italy Cost Action Short-term Scientific Missions.	Apr 2024
Participation in EuCAP 2023 – Florence, Italy FRS FNRS <i>Participation in a scientific meeting.</i>	Mar 2023

Personal Achievements

ADE Scheduler: An open source alternative to UCLouvain's scheduling tool, that became the official university tool in Sep. 2024.

Manim Slides: An open source presentation tool, used by many for teaching, conference presentations, or thesis defenses.

Scouts: 6 years as a scout (Les Scouts) leader.

Reviews

EuCAP: EuCAP 2026 (5 reviews)

IEEE: Access (5 reviews), Transactions on Vehicular Technology (7 reviews), Antennas and Wireless Propagation Letters (1 review), and Open Journal of the Communications Society (1 review)

Journal of Open Source Software: 1 review (see [#9384](#))

Nature Partner Journals: Wireless Technology: 1 review

Referees

Prof. Claude Oestges, UCLouvain: claud.oestges@uclouvain.be

Prof. Laurent Jacques, UCLouvain: laurent.jacques@uclouvain.be

Prof. Vittorio Degli-Esposti, University of Bologna: v.degliesposti@unibo.it